

Radicals Do Not Split Across a Sum

Answer Key

Algebra 2

Quick Answers

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|---|--|
| 1. (a) $\sqrt{9+16} = 5$, (b) $\sqrt{9}+\sqrt{16} = 7$, — | 5. (a) $\sqrt{75} = 5\sqrt{3}$, (b) $\sqrt{4/9} = \frac{2}{3}$ |
| 2. (a) $\sqrt{50} = 5\sqrt{2}$, (b) $\sqrt{72} = 6\sqrt{2}$ | 6. (a) the correct value at $x = 3 = 3.61$, (b) the student's expression at $x = 3 = 5$, — |
| 3. (a) left side at $x = 4 = 5$, (b) right side at $x = 4 = 7$, — | 7. (a) the correct solution = 4, — |
| 4. (a) $(36+64)$ to the one-half power = 10, (b) the two one-half powers added = 14 | 8. (a) exact simplified sum = $5\sqrt{3}$, (b) that sum as a decimal = 8.66, (c) the root of the sum, as a decimal = 6.24 |
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What is verified

16 of 20 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 3, 6, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.A.2 – *problem 7*

HSN-RN.A.2 – *problems 1, 2, 3, 4, 5, 6, 8*

Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

3. *If they got 7: split the root across the sum, writing the left side as $x + 3$ and getting 7*
6. *If they got 5: split the root across the sum and wrote $2 + x$, giving 5*
7. *If they got 16: merged the two radicals into a single root of the sum, solved $x + 9 = 25$ and reported 16*
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1. With 9 and 16: (a) $\sqrt{9+16}$. (b) $\sqrt{9} + \sqrt{16}$. (c) What the comparison shows.

Solution (a): $\sqrt{9+16} = \sqrt{25}$.

Solution (b): $\sqrt{9} + \sqrt{16} = 3 + 4$.

Solution (c): *Instructor-judged.* The two results are different, 5 against 7, so $\sqrt{a+b}$ and $\sqrt{a} + \sqrt{b}$ are not the same expression. One counterexample is enough to kill a proposed identity: taking a square root term by term across a sum is never valid. Full credit needs the general conclusion, not just the two numbers.

5

7

2. Simplest form: (a) $\sqrt{50}$. (b) $\sqrt{72}$.

Solution (a): Split off the largest perfect-square *factor* — this is a product, so the root does come apart: $\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{25}\sqrt{2}$. $5\sqrt{2}$

Solution (b): $\sqrt{72} = \sqrt{36 \cdot 2} = \sqrt{36}\sqrt{2}$. $6\sqrt{2}$

3. Student claim $\sqrt{x^2 + 9} = x + 3$. (a) Left side at $x = 4$. (b) Right side at $x = 4$. (c) The error.

Solution (a): $\sqrt{4^2 + 9} = \sqrt{16 + 9} = \sqrt{25}$. 5

Solution (b): $4 + 3$. 7

Solution (c): *Instructor-judged.* The student applied the product rule to a sum. $\sqrt{ab} = \sqrt{a}\sqrt{b}$ holds, but there is no rule that takes $\sqrt{a + b}$ apart, and the single test at $x = 4$ ($5 \neq 7$) settles it. $x^2 + 9$ is a sum of squares and does not factor over the real numbers, so $\sqrt{x^2 + 9}$ is already in simplest form — nothing can be brought out.

4. With 36 and 64: (a) $(36 + 64)^{1/2}$. (b) $36^{1/2} + 64^{1/2}$.

Solution (a): $(36 + 64)^{1/2} = 100^{1/2} = \sqrt{100}$. 10

Solution (b): $36^{1/2} + 64^{1/2} = 6 + 8$. 14

The exponent $\frac{1}{2}$ obeys the same rules as the radical sign: it distributes over a product, never over a sum.

5. Simplest form: (a) $\sqrt{75}$. (b) $\sqrt{4/9}$.

Solution (a): $\sqrt{75} = \sqrt{25 \cdot 3} = \sqrt{25}\sqrt{3}$. $5\sqrt{3}$

Solution (b): A quotient does split: $\sqrt{\frac{4}{9}} = \frac{\sqrt{4}}{\sqrt{9}} = \frac{2}{3}$. $\frac{2}{3}$

6. Student claim $\sqrt{4 + x^2} = 2 + x$. (a) Correct value at $x = 3$. (b) Student's value at $x = 3$. (c) The error.

Solution (a): $\sqrt{4 + 3^2} = \sqrt{4 + 9} = \sqrt{13} = 3.6055\dots$, which rounds to 3.61

Solution (b): $2 + 3$. 5

Solution (c): *Instructor-judged.* The student took the square root of each term separately, which the comparison $3.61 \neq 5$ rules out. Since $4 + x^2$ is a sum of two squares it has no real factorisation, so no factor can come out of the radical: $\sqrt{4 + x^2}$ is already simplest. Any alternative form a student offers should be tested at a number before it is accepted.

7. Student solved $\sqrt{x} + \sqrt{9} = 5$ as $\sqrt{x + 9} = 5$, $x = 16$. (a) Correct solution. (b) The error.

Solution (a): Evaluate the known radical first, then isolate:

$$\begin{aligned}\sqrt{x} + 3 &= 5 \\ \sqrt{x} &= 2 \\ x &= 4.\end{aligned}$$

Check: $\sqrt{4} + \sqrt{9} = 2 + 3 = 5$ ✓

$x = 4$

Solution (b): *Instructor-judged.* The first line replaces a sum of two radicals with the radical of the sum, which is the same invalid move as before. Testing the student's answer settles it: $\sqrt{16} + \sqrt{9} = 4 + 3 = 7$, not 5. A full-credit answer names that step and says the two radicals must be handled separately.

8. Challenge: $\sqrt{12} + \sqrt{27}$. (a) Exact simplified form. (b) As a decimal. (c) $\sqrt{12+27}$ as a decimal.

Solution (a): Simplify each radical separately, then collect like radicals:

$$\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3}.$$

Solution (b): $5\sqrt{3} = 5(1.73205\dots) = 8.6602\dots$

Solution (c): $\sqrt{12+27} = \sqrt{39} = 6.2449\dots$

$5\sqrt{3}$

8.66

6.24

The two decimals differ, which is the same lesson as problem 1: adding under the radical and adding the radicals give different numbers.